Smarter Serverless Autoscaling with Contextual Bandits

Moving beyond fixed rules with online decision-making.

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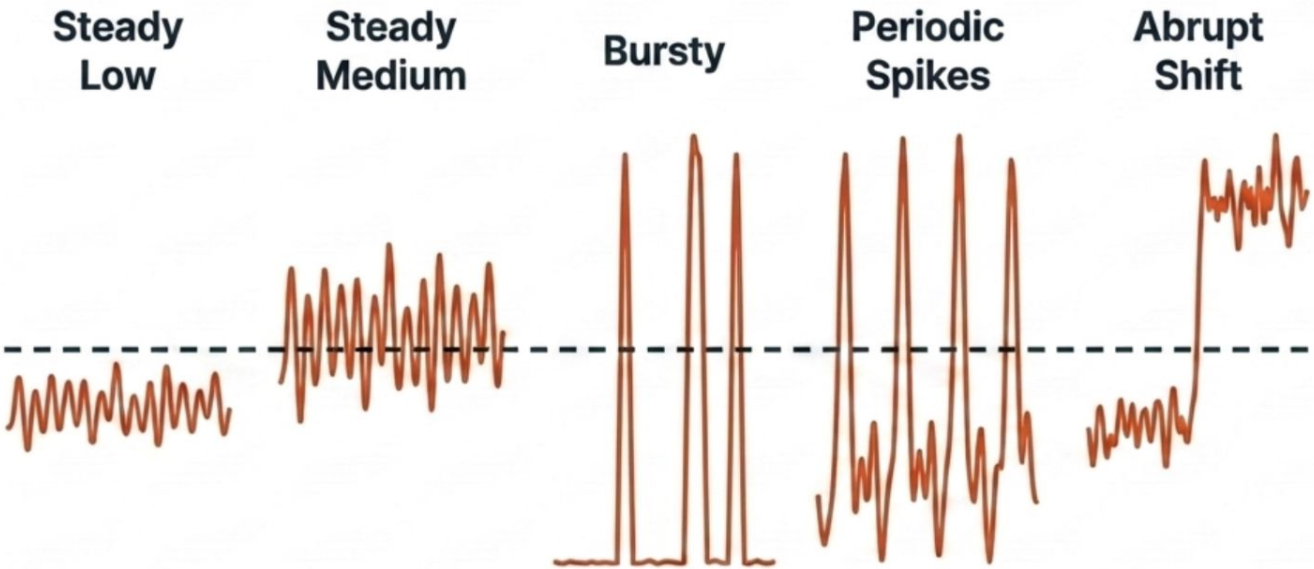
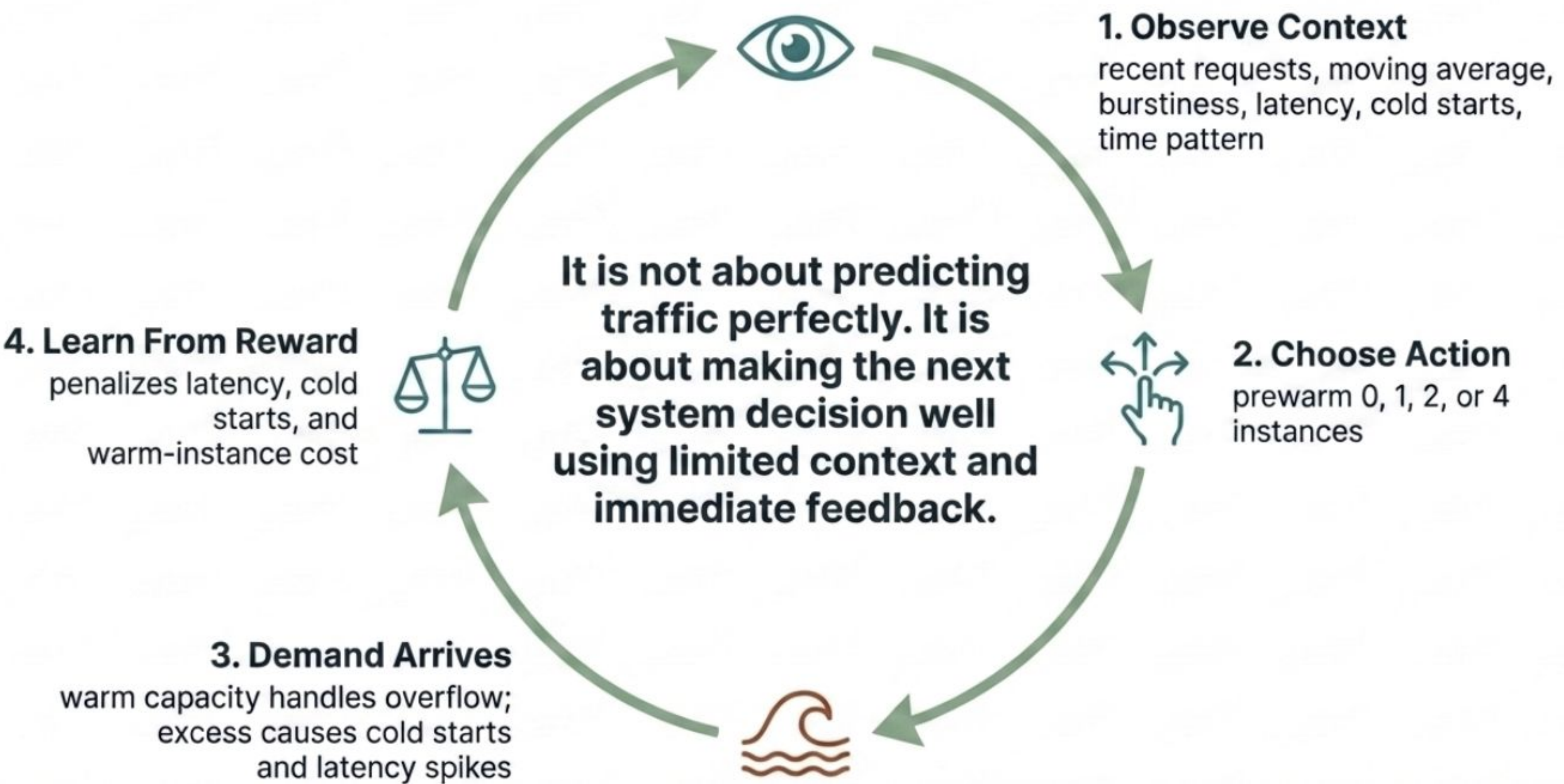
The Limit of Fixed Heuristics: Traffic Never Has Just One Shape



Serverless functions are cheap when demand is low, but unexpected spikes cause severe latency via cold starts.

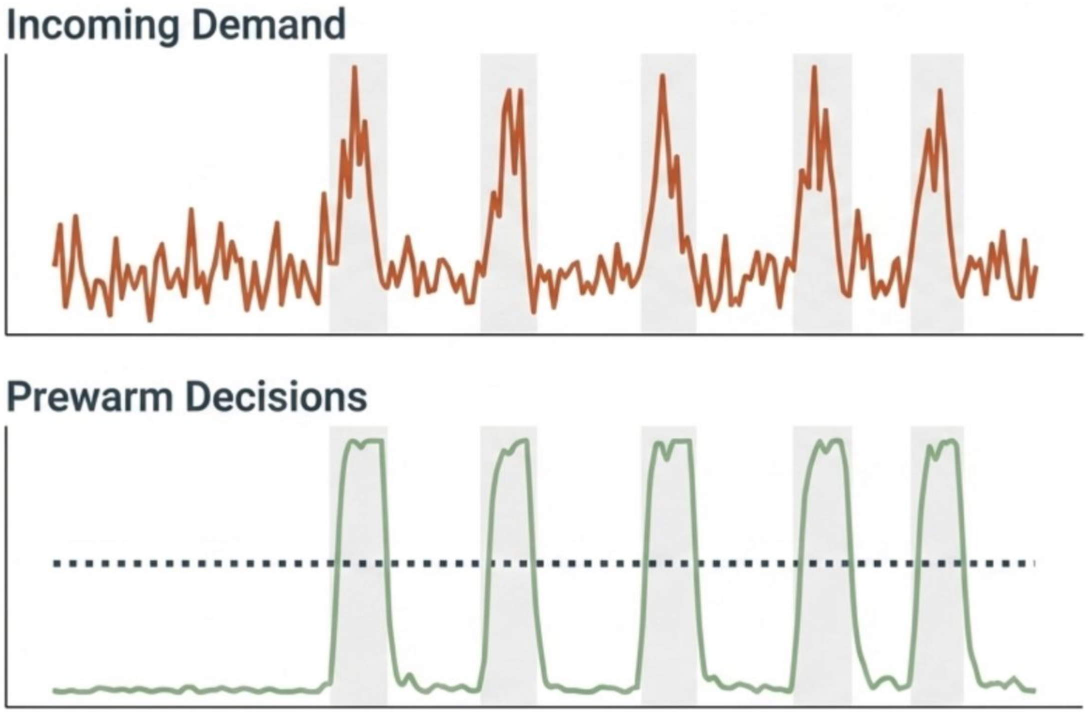
Fixed rules are simple, but they are inherently **brittle** when **traffic** changes.

The Mechanism: Online Decision Making Under Uncertainty



The Outcome: Sharply Cutting Cold Starts When It Matters

The agent learns the repeating pattern, sharply cutting cold starts during spikes.



Static Rule (Always 2)

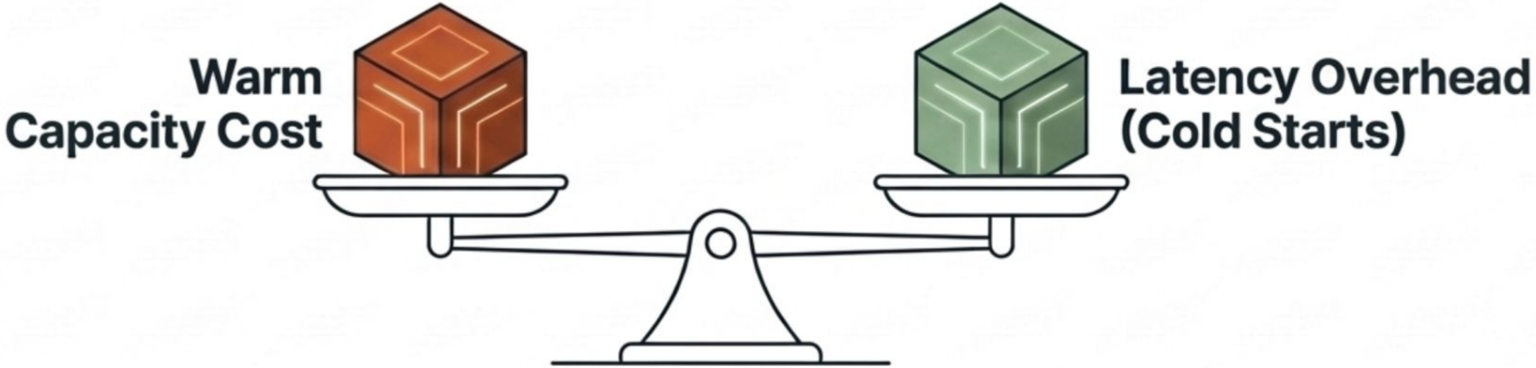
Cold starts: 450
Total cost: 48.0
Avg latency: 149.3 ms

Contextual Bandit (Linear Thompson Sampling)

Cold starts: 239
Total cost: 93.4
Avg latency: 140.0 ms

Data from seed 11, horizon 120

The Sweet Spot of Contextual Bandits



The Goal:

Minimize latency overhead without paying to keep too many instances warm.

The Pragmatic Reality:

The learner spends slightly more money during idle times to explore, but it learns repeating patterns fast enough to cut cold starts exactly when traffic spikes occur. **The latency benefit substantially exceeds the exploration cost.**

What the Policy Sees, Chooses, and Optimizes

Each time window, the model observes recent behavior, picks a prewarm level, then learns from the result.

CONTEXT (INPUTS)

What the model looks at before deciding:

- Recent request count
- Short moving average
- Burstiness / sudden spikes
- Recent cold starts
- Recent latency
- Time of day (optional)

decide

ACTION

Choose the next prewarm level

- 0 warm
- 1 warm
- 2 warm
- 4 warm

measure

REWARD (GOAL)

A good choice should keep all four of these low:

- **Latency — slow responses**
- **SLA violations — missed target**
- **Cold starts — startup delay**
- **Warm cost — extra idle capacity**

How We Evaluate the Idea

We compare bandit policies against simple rules under multiple traffic patterns and measure both speed and cost.

ALGORITHMS

Policies compared

- LinUCB
- Linear Thompson Sampling

Baselines

- Always 0 / 1 / 2 warm
- Threshold rule

WORKLOADS

Traffic patterns we simulate

- Steady low load
- Steady medium load
- Bursty traffic
- Periodic spikes
- Abrupt shifts

Why several workloads?

A method can look great on steady traffic and still fail when demand shifts suddenly.

METRICS

What success looks like

- Lower average latency
- Lower p95 latency
- Fewer SLA violations
- Lower total cost

Why This Matters in Practice

Adaptive prewarming helps serverless systems stay fast, affordable, and reliable under changing demand.

Performance

- Fewer cold starts
- Lower latency
- Fewer SLA misses

Smarter prewarming choices

Robustness

- Adapts when traffic is bursty
- Handles periodic spikes bette
- Useful when demand shifts

Cost

- Avoids extra idle capacity
- Reduces unnecessary warm cost